Dataset Description

Data Source:

The dataset used in this project was obtained from **Kaggle** specifically: [Gym Members Exercise Dataset]

This dataset contains information about gym members, their physical attributes, workout sessions, and performance metrics such as calories burned and heart rate.

Before Processing:

- The dataset was originally on a **single flat table (CSV file)**
- It contained mixed information:
  - Member's details
  - Workout details
  - Session performance

Steps Applied:

1. Data Cleaning

- Column names were standardized:
  - Converted to lowercase
  - Removed spaces and special characters
- Ensured consistent data types (numeric vs text)

2. Adding Unique Identifiers

- A unique member_id was added to identify each member
- A workout_id was created for workout types

3. Data Normalization

The dataset was split into multiple tables to simulate a real database design:

Members Table:

- Contains personal information:
    - member_id
    - first_name
    - gender
    - age
    - weight
    - fat_percentage

Workouts Table:

- Contains workout-related data:
    - workout_id
    - workout_type
    - experience_level

Sessions Table:

- Contains performance data:
    - member_id
    - workout_id
    - calories_burned
    - session_duration_hours
    - avg_bpm
    - max_bpm
    - resting_bpm

This transformation improves:

- Data organization
- Query performance
- Real-world database simulation

4. Data Enrichment

To make the dataset more realistic:

- A new column first_name was added
- Names were generated randomly based on gender:
  - Male → Ahmed, Mohamed, Omar, etc.
  - Female → Sara, Mona, Nour, etc.

This step improves:

- Readability of results
- Real-world simulation

5. Data Organization in Excel

- The dataset was initially handled in **Excel (.xlsx)**
- Split into multiple sheets:
  - Members
  - Workouts
  - Sessions
- Then exported as **CSV files** for SQL Server import

Why These Steps Matter

- Simulates real-world database systems
- Enables advanced SQL operations:
  - Aggregations
  - Subqueries
  - Multi-table analysis
- Improves clarity and scalability of the data

```sql
/* 3dd kol gender*/
select  gender , count(*) as total
from members
group by gender
```

	gender	total
1	Male	511
2	Female	462

Insight: Helps identify if the gym audience is male-dominated or balanced.

```sql
/*avg age l kol gender */
select gender , avg (age ) as avg_age
from members
group by gender
```

	gender	avg_age
1	Male	38
2	Female	38

Insight: Shows which gender group is generally younger or older.

```sql
/*top 15 mn na7yt el weight*/
select top 15 id, name , weight , fat_percentage
from members
order by weight desc
```

	id	name	weight	fat_percentage
1	594	Youssef	129.899993896484	21.7000007629395
2	123	Youssef	129.5	25.2000007629395
3	855	Ali	129.199996948242	21.3999996185303
4	97	Mohamed	129	27.1000003814697
5	284	Omar	128.399993896484	26.1000003814697
6	292	Ahmed	128.399993896484	20.3999996185303
7	615	Mostafa	128.399993896484	23.6000003814697
8	181	Mostafa	128.199996948242	26.2000007629395
9	649	Ahmed	127.900001525879	21.7999992370605
10	869	Ali	127.800003051758	28.7000007629395
11	346	Karim	127.699996948242	24.1000003814697
12	36	Ali	127.599998474121	27.2999992370605
13	816	Omar	127.5	29.7999992370605
14	850	Youssef	127.099998474121	24.8999996185303
15	946	Youssef	127.099998474121	23.2999992370605

Insights: Useful for identifying potential weight-loss target users.

```sql
/*top 15 body fat*/
select top 15 id,name,fat_percentage , weight
from members
order by fat_percentage desc
```

	id	name	fat_percentage	weight
1	624	Nour	35	65.4000015258789
2	781	Sara	35	72.8000030517578
3	490	Sara	34.90000152...	54.2000007629395
4	538	Sara	34.90000152...	42.7000007629395
5	554	Sal...	34.90000152...	70.6999969482422
6	676	Farah	34.90000152...	63
7	680	Sara	34.90000152...	75.5999984741211
8	924	Mona	34.90000152...	70.5999984741211
9	266	Sara	34.79999923...	63.0999984741211
10	332	Sara	34.79999923...	63.5999984741211
11	492	Aya	34.79999923...	44.0999984741211
12	580	Laila	34.79999923...	72.5999984741211
13	42	Sal...	34.70000076...	44.2999992370605
14	291	Aya	34.70000076...	42.9000015258789
15	315	Laila	34.70000076...	63.0999984741211

Insights: Useful for identifying members who may need specialized fitness or nutrition programs.

Insights: Helps identify the dominant age group in the gym.

	age_group	total
1	older	463
2	adult	345
3	young	165

```sql
/*age slicer*/
select
    case
        when age < 25 then 'young'
        when age > 40 then 'older'
        else 'adult'
    end as age_group,
    count(*) as total
from members
group by
    case
        when age < 25 then 'young'
        when age > 40 then 'older'
        else 'adult'
    end
order by total desc;
```

insight: identify the workout different types

```sql
select distinct (workout_type)
from workouts
```

	workout_type
1	Yoga
2	Strength
3	HIIT
4	Cardio

```sql
/*best workout for burn calories*/
select distinct( w.workout_type) , avg ( s.calories_burned) as avg_calories_burned
from sessions as s , workouts as w
where w.id = s.workout_id
group by w.workout_type
order by avg_calories_burned desc
```

	workout_type	avg_calories_burned
1	HIIT	925
2	Strength	910
3	Yoga	903
4	Cardio	884

Insight: Workout types with higher averages are more intense and effective for fat loss.

```sql
/*longest session l kol type*/
select w.workout_type , max (s.session_duration) as session_duration
from workouts as w , sessions as s
where w.id = s.workout_id
group by w.workout_type
```

	workout_type	session_duration
1	Yoga	1.99000000953674
2	Strength	2
3	HIIT	1.99000000953674
4	Cardio	1.99000000953674

Insight: longest session for every workout type

```sql
/*max. HR */
select name , (select max(max_bpm) from sessions ) as bpm
from members
where id in (
    select members_id
    from sessions
    where max_bpm = (select max(max_bpm) from sessions )
    )
```

	name	bpm
1	Nour	199
2	Laila	199
3	Farah	199
4	Karim	199
5	Ahm...	199
6	Nour	199
7	Mos...	199
8	Aya	199
9	Ali	199
10	Karim	199
11	Ahm...	199

Insight: identify the names of the maximum heart rate

```sql
/*avg water intake l kol type*/
select w.workout_type , avg (s.water_intake) as water_intake
from workouts as w , sessions as s
where w.id = s.workout_id
group by w.workout_type
order by water_intake desc
```

	workout_type	water_intake
1	HIIT	2.65294117679423
2	Yoga	2.64225941821621
3	Cardio	2.61686274724848
4	Strength	2.59922480721806

Insight: which workout type uses more water than others

```sql
/*highest 10 burn calories*/
select top 10 m.name , s.calories_burned , w.workout_type
from members as m , sessions as s , workouts as w
where m.id = s.members_id and s.workout_id = w.id
order by calories_burned desc
```

Insights: best 10 people burning calories

	name	calories_burned	workout_type
1	Karim	1783	Strength
2	Ahmed	1766	HIIT
3	Omar	1725	Strength
4	Youssef	1701	Strength
5	Mostafa	1688	HIIT
6	Karim	1675	Strength
7	Karim	1646	HIIT
8	Mohamed	1634	Strength
9	Laila	1625	Cardio
10	Mostafa	1622	HIIT

```sql
/*avg burning vs experience level*/
select experience_level ,
    (select avg (calories_burned) from sessions s where s.members_id in
        (select id from members where experience_level = m.experience_level )) as avg_calorie
from members m
group by experience_level
```

	experience_level	avg_calorie
1	3	1265
2	1	726
3	2	901

```sql
/*Elite people*/
select name , experience_level
from members
where id in (select members_id
            from sessions
            where avg_bpm >150
                and calories_burned > 1000
                and resting_bpm < 65)
```

	name	experience_level
1	Mohamed	3
2	Youssef	2
3	Ali	2
4	Laila	2
5	Mona	1
6	Laila	2
7	Ali	1
8	Youssef	2
9	Karim	1
10	Nour	2
11	Mohamed	1
12	Hassan	2
13	Hana	2
14	Farah	3
15	Mostafa	3
16	Farah	3
17	Ahmed	1
18	Laila	3
19	Hana	3